Adjective Comparison Across Random Seeds

Subsamples of 53 rows, with seeds 42, 1, 11, and 13.

Main idea. The results are very consistent across random subsamples. No matter the seed, the two datasets remain clearly different in adjective usage, while Dataset B keeps a broader vocabulary.

Observed ranges

Cosine similarity	0.3352 – 0.3444
Jensen–Shannon distance	0.6652 – 0.6706
Entropy A	6.7146 (constant)
Entropy B	6.7442 – 6.8743
Shared adjectives	111 – 120
A-only adjectives	101 – 110
B-only adjectives	152 – 179
Mean row-wise Jaccard	0.046 – 0.053
Mean row-wise overlap	1.755 – 2.075

Interpretation

The stability across seeds suggests that the overall conclusion is robust to random down-sampling. Cosine similarity stays low, Jensen–Shannon distance stays high, and row-wise Jaccard remains very small, so the two groups do not describe the ads in closely matching adjective terms. Dataset B also consistently contains more unique adjectives and slightly higher entropy.

In short: even after repeated random subsampling, the lexical gap between the two datasets remains strong and stable.

What the scores mean

- **Cosine similarity:** compares the overall pattern of adjective frequencies. Higher means the two datasets emphasize similar adjectives.
- **Jensen–Shannon distance:** measures how far apart the two adjective distributions are. Lower means more similar; higher means more different.
- **KL divergence** $(A \to B)$ **and** $(B \to A)$**:** measures how badly one distribution approximates the other. A value of ∞ means some adjectives appear in one dataset but not at all in the other.
- **Entropy A / Entropy B:** measures vocabulary diversity within each dataset. Higher entropy means adjective usage is more varied and less concentrated on a few words.
- **Shared adjectives:** the number of adjective lemmas appearing in both datasets.
- **A-only / B-only adjectives:** the number of adjective lemmas unique to one dataset.
- **Mean row-wise Jaccard:** average proportion of shared adjectives within paired responses. Low values mean little direct overlap.
- **Mean row-wise overlap:** average number of shared adjectives per paired response.
- **`n_a_total` / `n_b_total`:** average number of extracted adjectives per response in each dataset.

Additional text similarity results

- **TF-IDF centroid cosine:** 0.6200
- **Mean row-wise TF-IDF cosine:** 0.1550

The centroid-level TF-IDF similarity is moderately high, which suggests that, at a broad corpus level, the two datasets still revolve around related themes and ad-related vocabulary. However, the mean row-wise TF-IDF similarity is low, indicating that matched responses remain quite different in their specific wording and local emphasis.

Lexical richness

- **Dataset A mean token count:** 137.87
- **Dataset B mean token count:** 470.70
- **Dataset A TTR:** 0.6302
- **Dataset B TTR:** 0.4594
- **Dataset A distinct-2:** 0.8945
- **Dataset B distinct-2:** 0.8815
- **Dataset A hapax ratio:** 0.7123
- **Dataset B hapax ratio:** 0.6620

Dataset B produces far longer responses and many more total adjective tokens, which confirms that it is much more verbose. Dataset A nevertheless has the higher type-token ratio and hapax ratio, meaning its shorter responses are relatively denser in unique vocabulary, whereas Dataset B repeats more within its much larger output.

SSR-style semantic rating summary

- **SSR mean expected score, A:** 2.9449
- **SSR mean expected score, B:** 2.7671
- **SSR KS similarity:** 0.9337
- **SSR Wasserstein distance:** 0.1779
- **SSR cosine similarity:** 0.9904

When the free-text answers are mapped onto a latent 5-point semantic scale, the two datasets look much closer than they do at the raw lexical level. This suggests that although the exact adjectives differ strongly, the underlying semantic orientation of the responses is still broadly similar, with Dataset A appearing slightly more positive on average than Dataset B.

Interpretation

Taken together, the results show a clear distinction between *lexical divergence* and *semantic proximity*. At the word level, overlap remains low: cosine similarity is only 0.3352, Jensen–Shannon distance is 0.6706, and the mean row-wise Jaccard overlap is just 0.0527. At the same time, the SSR-style mapping and TF-IDF centroid similarity suggest that both datasets still express broadly related reactions to the ads, even if one dataset does so with much longer and more varied language.

In short: the datasets differ strongly in wording, length, and surface vocabulary, but remain much closer in their broader semantic signal than raw lexical metrics alone would suggest.

Orange vs. Bouygues within the same dataset: AI Agents

Analysis settings

- **Word type:** adjectives
- **Rows analysed:** 800
- **Bouygues columns:** `Q_bouygues_memorisation_1`, `Q_bouygues_resonance_emotionnelle_1`, `Q_bouygues_image_1`, `Q_bouygues_intention_achat_1`
- **Orange columns:** `Q_orange_memorisation_1`, `Q_orange_resonance_emotionnelle_1`, `Q_orange_image_1`, `Q_orange_intention_achat_1`
- **SSR alpha:** 0.0
- **SSR temperature:** 1.0
- **Reference sets:** 3

This comparison is performed *within the same dataset*, meaning that Orange and Bouygues are compared across responses from the same pool of respondents.

Lexical distribution scores

- **Cosine similarity:** 0.3842
- **Jensen–Shannon distance:** 0.6169
- **Entropy (Bouygues):** 6.7039
- **Entropy (Orange):** 6.2590
- **Smoothed KL Bouygues → Orange:** 3.1937
- **Smoothed KL Orange → Bouygues:** 2.9452

At the adjective-distribution level, Orange and Bouygues are clearly *not* lexically identical. The cosine similarity remains moderate, while the Jensen–Shannon distance is relatively high, indicating that the two brands elicit substantially different adjective profiles.

Vocabulary overlap

- **Shared adjectives:** 246
- **Bouygues-only adjectives:** 200
- **Orange-only adjectives:** 166
- **Bouygues vocabulary size:** 446
- **Orange vocabulary size:** 412
- **Bouygues adjective tokens:** 10,348
- **Orange adjective tokens:** 9,510

There is a substantial shared core vocabulary, but each brand also has a large set of exclusive adjectives. Bouygues appears slightly broader in adjective usage, both in total vocabulary size and in total adjective token count.

Row-wise overlap

- **Mean Jaccard overlap:** 0.0632
- **Median Jaccard overlap:** 0.0526
- **Maximum Jaccard overlap:** 0.3750
- **Mean shared adjectives per row:** 1.2888
- **Mean Bouygues-only adjectives per row:** 11.0613
- **Mean Orange-only adjectives per row:** 9.5713
- **Mean total Bouygues adjectives per row:** 12.9350
- **Mean total Orange adjectives per row:** 11.8875

At the respondent level, overlap is low. On average, the same respondent uses only about 1.29 shared adjectives across both brands, while using around 10–11 brand-specific adjectives on each side, which suggests that respondents describe Orange and Bouygues in quite distinct lexical terms.

Textual similarity

- **TF-IDF centroid cosine:** 0.6866
- **Mean row-wise TF-IDF cosine:** 0.1259

The corpus-level TF-IDF centroid cosine is moderately high, which means the two brands still live in a related global semantic space. However, the low row-wise TF-IDF cosine shows that, for a given respondent, the exact textual realization differs strongly between Bouygues and Orange.

Lexical richness

- **Bouygues mean token count:** 167.21
- **Orange mean token count:** 166.82
- **Bouygues TTR:** 0.6481
- **Orange TTR:** 0.6250
- **Bouygues distinct-2:** 0.9505
- **Orange distinct-2:** 0.9444
- **Bouygues hapax ratio:** 0.7344
- **Orange hapax ratio:** 0.7399

The two brands generate very similar response lengths overall. Bouygues is slightly richer in type-token ratio and distinct bigrams, while Orange has a marginally higher hapax ratio, so lexical richness is broadly comparable with only small differences in favor of Bouygues on variety and Orange on rare-word usage.

SSR-style semantic scores

- **SSR mean expected score, Bouygues:** 2.7004
- **SSR mean expected score, Orange:** 2.7617
- **SSR KS similarity:** 0.9760
- **SSR Wasserstein distance:** 0.0614
- **SSR cosine similarity:** 0.9976

When mapped to a latent 5-point semantic scale, the two brands become extremely close. The very high KS similarity and cosine similarity, together with the very low Wasserstein distance, indicate that Orange and Bouygues produce almost the same *distribution of semantic orientation*, even though they differ much more at the raw lexical level.

SSR distributions

$$Bouygues = 0.27440.28920.07180.19080.1738$$

$$Orange = 0.25030.30490.06660.18900.1891$$

Both distributions place most probability mass on the lower-middle part of the scale, with relatively similar shapes. Orange is slightly shifted upward compared with Bouygues, which is consistent with its slightly higher SSR mean expected score.

Interpretation

These results point to a strong contrast between *surface-level wording* and *underlying semantic position*. At the lexical level, Orange and Bouygues are clearly differentiated: row-wise overlap is low, Jensen–Shannon distance is high, and exact matched responses are not especially similar. At the semantic-distribution level, however, the two brands are extremely close under the SSR mapping, suggesting that respondents use different adjectives to talk about the two brands while still expressing broadly similar latent evaluative signals.

1 Human respondents: Orange vs. Bouygues

Analysis settings

- **Word type:** adjectives
- **Rows analysed:** 53
- **Bouygues columns:** `Q_bouygues_memorisation_1`, `Q_bouygues_resonance_emotionnelle_1`, `Q_bouygues_image_1`, `Q_bouygues_intention_achat_1`
- **Orange columns:** `Q_orange_memorisation_1`, `Q_orange_resonance_emotionnelle_1`, `Q_orange_image_1`, `Q_orange_intention_achat_1`
- **SSR alpha:** 0.0
- **SSR temperature:** 1.0
- **Reference sets:** 3

This analysis compares the Orange and Bouygues descriptions produced by the human respondents only.

Lexical distribution scores

- **Cosine similarity:** 0.5208
- **Jensen–Shannon distance:** 0.7083
- **Entropy (Bouygues):** 6.5536
- **Entropy (Orange):** 6.2443
- **Smoothed KL Bouygues → Orange:** 10.8454
- **Smoothed KL Orange → Bouygues:** 8.6645

At the adjective-distribution level, the human descriptions of Orange and Bouygues are markedly different. The cosine similarity is only moderate, while the Jensen–Shannon distance is high, indicating strong divergence in the adjective profiles associated with the two brands.

Vocabulary overlap

- **Shared adjectives:** 46
- **Bouygues-only adjectives:** 79
- **Orange-only adjectives:** 51
- **Bouygues vocabulary size:** 125
- **Orange vocabulary size:** 97
- **Bouygues adjective tokens:** 234
- **Orange adjective tokens:** 182

The shared adjective core is relatively small, and Bouygues also attracts a broader exclusive vocabulary than Orange. This suggests that human respondents use more varied adjective choices when describing Bouygues than when describing Orange.

Row-wise overlap

- **Mean Jaccard overlap:** 0.0459
- **Median Jaccard overlap:** 0.0000
- **Maximum Jaccard overlap:** 0.6667
- **Mean shared adjectives per row:** 0.4528
- **Mean Bouygues-only adjectives per row:** 3.6792
- **Mean Orange-only adjectives per row:** 2.5849
- **Mean total Bouygues adjectives per row:** 4.4151
- **Mean total Orange adjectives per row:** 3.4340

At the respondent level, overlap is extremely low. The median Jaccard score is exactly 0, meaning that for at least half of the human respondents, the adjective sets used for Bouygues and Orange do not overlap at all.

Textual similarity

- **TF-IDF centroid cosine:** 0.6445
- **Mean row-wise TF-IDF cosine:** 0.1066

At the corpus level, the two brands still occupy a moderately similar textual space. However, the very low row-wise TF-IDF cosine again shows that individual respondents tend to use substantially different wording when describing Bouygues versus Orange.

Lexical richness

- **Bouygues mean token count:** 51.28
- **Orange mean token count:** 44.62
- **Bouygues TTR:** 0.8236
- **Orange TTR:** 0.8450
- **Bouygues distinct-2:** 0.9795
- **Orange distinct-2:** 0.9569
- **Bouygues hapax ratio:** 0.8459
- **Orange hapax ratio:** 0.8676

Human responses are short but lexically very dense, with extremely high type-token ratios and hapax ratios. Orange appears slightly more lexically concentrated in unique-word usage, while Bouygues is slightly longer and somewhat richer in distinct bigrams.

SSR-style semantic scores

- **SSR mean expected score, Bouygues:** 2.9042
- **SSR mean expected score, Orange:** 2.9964
- **SSR KS similarity:** 0.9666
- **SSR Wasserstein distance:** 0.0922
- **SSR cosine similarity:** 0.9963

Despite the strong lexical divergence, the SSR-style mapping places the two human brand-response distributions extremely close to one another. Orange is slightly shifted upward relative to Bouygues, but the overall semantic distributions remain highly similar.

SSR distributions

$$Bouygues = 0.22700.26720.10250.18130.2221$$

$$Orange = 0.19360.26860.11130.20080.2257$$

Both distributions have similar shapes, but Orange places slightly less mass on the lower end of the scale and slightly more mass on the upper end. This is consistent with its marginally higher SSR mean expected score.

Interpretation

For the human subset, Orange and Bouygues are clearly separated at the lexical level but remain very close at the semantic-distribution level. In other words, human respondents tend to use different adjectives for the two brands, yet those differences do not translate into strongly different latent evaluative profiles under the SSR mapping.